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SECTION 47

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

COURSE NAME

OBJECT ORIENTED PROGRAMMING USING

JAVA

COURSE CODE (R1UC201C)

B.TECH. SECOND SEM (2024 -2025)

PROJECT TITLE: ONSLAUGHT (VISAGE VENGEANCE)

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(OOPS USING JAVA)

INTRODUCTION

ONSLAUGHT

“VISAGE VENGEANCE”

"Onslaught: Visage Vengeance" is a **Java-based 2D action shooter game** that combines futuristic gameplay with real-time face recognition for secure login and personalized experience.

Set in a dystopian world where identity is power, players take on the role of a resistance fighter battling through hostile environments to uncover a deep conspiracy. The game features smooth player controls, dynamic enemy waves, and an engaging progression system.

🔧 **Key Features:**

- 🎮 **Built with JavaFX** for user interface elements like menus, login, and profile selection.
- 🔫 **Game Engine using LibGDX** for high-performance 2D graphics and game logic.
- 🧠 **Face Recognition Login** using OpenCV and deep learning (JavaCV) for secure and immersive access.
- 🧱 Modular code architecture for easy expansion and maintenance.
- 🎨 Pixel-art style environment with layered parallax effects and responsive sound design.

PRIMARY USES OF THE ONSLAUGHT (VISAGE VENGEANCE)

🎮 GAME DEVELOPMENT WITH JAVA:
This project showcases the use of Java for creating a dynamic 2D action game. With JavaFX for UI and LibGDX for game mechanics, it demonstrates Java's capabilities in real-time game development.

🧠 FACE RECOGNITION INTEGRATION:
The game integrates face recognition for login using JavaCV and OpenCV, offering secure authentication and a personalized experience for each player.

📚 EDUCATIONAL RESOURCE FOR GAME DEVELOPMENT:
This project serves as an educational tool for learning game development, real-time mechanics, and integrating biometric authentication. It helps developers understand the application of game physics, collision detection, and user interaction.

🔐 BIOMETRIC SECURITY IN GAMING:
Face recognition as a login method demonstrates how biometric security can be incorporated into games, adding layers of personalization and security.

💻 SHOWCASING JAVA'S VERSATILITY:
The project highlights Java's ability to handle both UI design and complex game mechanics, proving its use in performance-critical applications like games.

🕵 ENHANCING USER ENGAGEMENT:
By using face recognition, the game adapts to each player's identity, enhancing user experience and fostering deeper engagement with the game world.

🚀 EXPANDING JAVA'S APPLICATION SCOPE:
"Onslaught" demonstrates how Java can be applied in gaming, breaking out of traditional desktop applications into the realm of interactive, real-time games.

⚔ INNOVATIVE GAMEPLAY MECHANICS:
The unique face recognition feature introduces a new level of interactivity, making the player's identity a central part of the gaming experience.

PROJECT REQUIREMENTS FOR

"ONSLAUGHT: (VISAGE VENGEANCE)"

1. HARDWARE REQUIREMENTS:

- **Processor:** Intel Core i3 or higher (or equivalent AMD processor)
- **RAM:** Minimum 4 GB (8 GB recommended)
- **Storage:** 1 GB of free disk space for game installation and asset
- **Graphics:** Dedicated graphics card with at least 1 GB VRAM (e.g., NVIDIA GeForce or AMD Radeon)
- **Camera:** For face recognition, a **webcam** with at least 720p resolution is required

2. SOFTWARE REQUIREMENTS:

- **Operating System:** Windows 10/11, macOS, or Linux
- **Java Development Kit (JDK):** Version 11 or higher
- **Integrated Development Environment (IDE):** Eclipse, IntelliJ IDEA, or NetBeans
- **Libraries and Frameworks:**
  - **LibGDX:** For game mechanics and graphics rendering
  - **JavaFX:** For the user interface and controls
  - **JavaCV/OpenCV:** For face recognition integration
- **Database:** SQLite or similar (optional for storing user profiles and game data)

3. DEVELOPMENT TOOLS:

- **Version Control:** Git and GitHub for version tracking and collaboration
- **Build Tools:** Apache Maven or Gradle for project dependencies and builds
- **Image Editing Software:** For assets like sprites and textures (e.g., Adobe Photoshop, GIMP)
- **Audio Editing Software:** For sound effects and music (e.g., Audacity, FL Studio)

4. NETWORK REQUIREMENTS:

- **Internet Connection:** Required for downloading dependencies, libraries, and updates

5. ADDITIONAL REQUIREMENTS:

- **Camera Calibration:** A system for configuring and testing face recognition before gameplay
- **Testing and Debugging Tools:** Unit testing frameworks (e.g., JUnit) for Java, visual testing tools for the UI
- **Game Asset Management System:** To organize and manage game assets (images, sounds, and animations) during development
- **Performance Optimization Tools:** Profiling tools (e.g., VisualVM) to monitor the performance of the game and optimize for smooth gameplay.

6. SECURITY AND PRIVACY:

- **Data Privacy Management:** Ensure user data (especially facial data) is securely handled and complies with data protection regulations (e.g., GDPR).
- **Encryption:** For face recognition data storage and transmission to maintain privacy and security.

7. USER DOCUMENTATION:

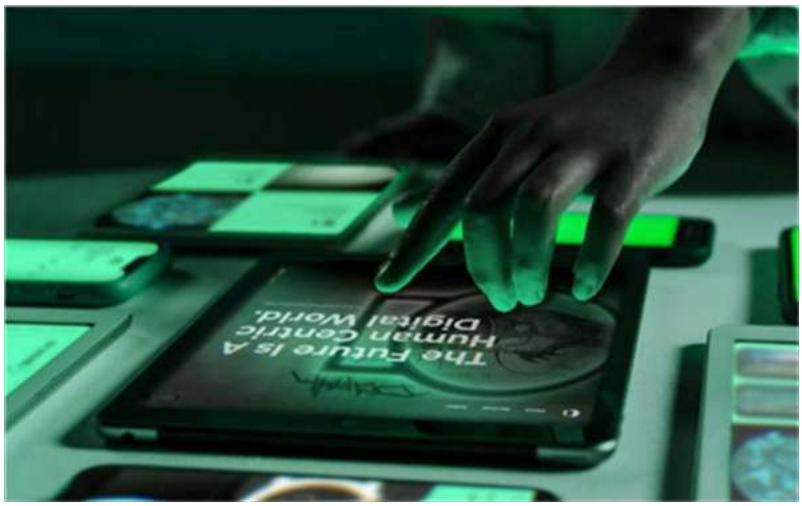
- **Help and Tutorial System:** An in-game guide or manual explaining how to use the face recognition login system and controls.
- **User Feedback Mechanism:** A system for collecting user feedback to improve gameplay and usability.

COMPONENTS OF THE "ONSLAUGHT: VISAGE VENGEANCE"

USER INTERFACE (UI):

The **UI** component is responsible for providing an interactive experience for the player. Using **JavaFX**, this component handles elements like the **main menu**, **settings screen**, **profile selection**, and **face recognition login**. The UI ensures smooth navigation and user-friendly interactions throughout the game.

- **Login Screen**: Integration of **face recognition** for user authentication
- **Menu System**: Main menu, pause menu, and settings options
- **HUD (Heads-Up Display)**: Displays vital game stats such as health, ammo, and score.



GAME ENGINE (LIBGDX):

The **game engine** is responsible for the core game mechanics. Built using **LibGDX**, this component handles the game's **physics engine**, **collision detection**, **input management**, and **sprite rendering**. It allows for real-time gameplay with smooth animations and interactions.

- **Game Loop**: The continuous cycle that updates the game state and renders the game's visuals
- **Player Controls**: Movement, shooting, and interacting with the environment
- **Enemy AI**: Behavior patterns for enemies that create challenges for the player

FACE RECOGNITION SYSTEM:

This is a key feature of the game, utilizing **JavaCV** and **OpenCV** libraries to implement **face recognition** technology. The system authenticates the player's identity during the login process, enhancing security and providing a personalized experience.

- **Face Detection**: Capturing the player's face using the webcam
- **Face Matching**: Comparing the captured face with stored data for authentication
- **Profile Linking**: Linking the recognized face with the player's saved game profile

GAME ASSETS:

Game assets include all the visual and auditory elements required for the game. These assets will be managed and optimized for performance using various editing tools and asset management systems.

- **Visual Assets**: Sprites, backgrounds, animations, and textures
- **Audio Assets**: Background music, sound effects for actions like shooting and explosions
- **Lighting and Effects**: Special effects like particle systems, lighting, and shadows to enhance visual appeal.

DATABASE/STORAGE SYSTEM:

The game needs to store player data, including profiles, progress, and settings. A simple **database** (like **SQLite**) or a **file-based storage system** will be used to persist these details

- **Player Profiles**: Storing player preferences, stats, and progress
- **Game Settings**: Saving and loading sound, difficulty, and display settings
- **High Scores and Achievements**: Tracking player milestones an rankings

MULTIPLAYER (OPTIONAL):

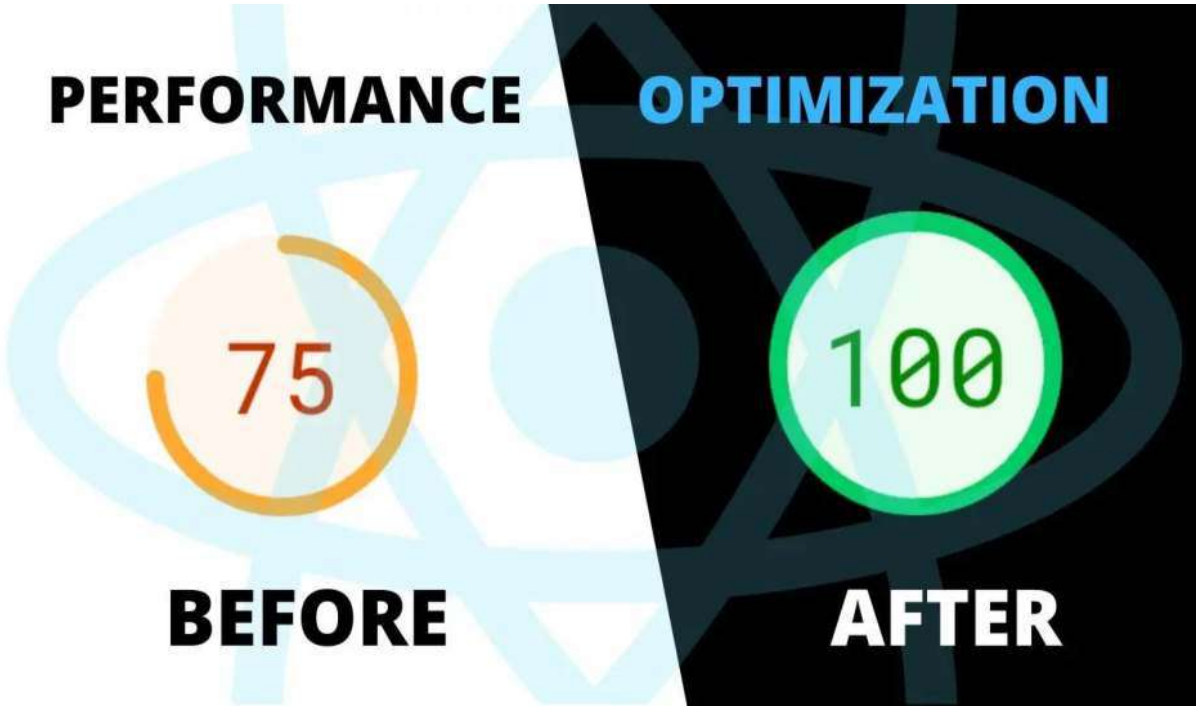
If multiplayer features are added, this component will handle **network communication** between players. It will manage the connection, synchronization of player states, and interaction within the game.

- **Server Management**: Handling multiple connections and data exchange
- **Matchmaking System**: Connecting players to multiplayer sessions

PERFORMANCE OPTIMIZATION:

This component is responsible for ensuring the game runs smoothly across various systems. **Profiling tools** (such as **VisualVM**) and **optimization techniques** will be used to ensure minimal latency and resource consumption.

- **Memory Management**: Efficient handling of assets and in-gam objects to avoid performance drops
- **Frame Rate Optimization**: Ensuring consistent frame rates for smooth gameplay
- **CPU/GPU Utilization**: Monitoring and optimizing the game's use of system resources



SECURITY AND PRIVACY:

As face recognition data is being used, the game needs a **secure system** for data storage and transmission to protect user privacy. Encryption and secure handling of personal data will be a priority.

- **Data Encryption**: Encrypting face data and personal informatio to protect user privacy
- **Secure Data Handling**: Ensuring compliance with data protection regulations like GDPR

CONCLUSION

"Onslaught: Visage Vengeance" is more than just a game—it is a **technological demonstration** of how traditional programming languages like **Java** can evolve to support **modern interactive applications**. The successful integration of **face recognition** not only enhances the **security and personalization** of the game but also opens doors for future applications in other industries like **edtech**, **healthcare**, and **authentication systems**.

This project has helped showcase:

- The **versatility of Java** in combining user interfaces, real-time processing, and advanced features.
- How **game development** can be a powerful educational tool for learning programming, AI, and UI design.
- The future potential of **biometric technology** in enhancing user experience and data protection.

By bridging **gaming and facial recognition**, this project sets an example of innovation in digital interaction, user identity, and immersive design.

OBJECTIVE

The primary objective of **"Onslaught: Visage Vengeance"** is to design and develop a **2D action shooter game** using Java that integrates **face recognition technology** for secure and personalized user login. This project aims to combine **biometric authentication** with interactive game mechanics to provide a **futuristic and immersive gaming experience**.

Key goals include:

- Demonstrating **Java's capability** in handling real-time game development using frameworks like **LibGDX** and **JavaFX**.
- Implementing **face detection and recognition** using **JavaCV/OpenCV** for secure, user-specific login.
- Creating engaging game elements such as **player movement**, **enemy AI**, and **HUD systems**.
- Offering students and developers a practical understanding of **game architecture**, **UI/UX design**, and **security integration**.
- Highlighting the potential of **Java in modern application areas** beyond traditional software, including entertainment and biometric tech.